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Bio-Organic Material Detection using Hydroxyapatite Composite QCM Sensors

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ABSTRACT

A hydroxyapatite (HAp)-coated quartz crystal microbalance (QCM) has been developed and used for bio-organic material detection. Polycrystalline HAp thin coatings were prepared on the gold-electrode of quartz crystals by hydrothermal post-annealing at 433 K from amorphous calcium phosphate films obtained by the rf-sputtering method. The shift in resonant frequency and the dissipation factor were simultaneously calculated from the measured decay curves. The QCM system found that bovine serum albumin (BSA) was immediately adsorbed on the HAp surfaces. The linear decrease in the dissipation factor suggested that the linking strength of HAp-BSA, BSA-BSA and/or BSA-liquid was slowly changed. It was implied that the HAp composite QCM was effective for the detection of bio-organic materials.

INTRODUCTION

Hydroxyapatite (HAp) ceramics possess a superior and selective adsorbability in reference to various bio-organic materials and have been applied to chromatographic columns for the isolation of nucleic acids, enzymes, polypeptides and viruses. A considerable number of thermodynamic and biochemical studies of the unique adsorption properties have revealed the preferential selectivity and mechanisms for each of the adsorbates [1-3]. A quartz crystal microbalance (QCM) is a sensitive weighing device [4]. The QCM can be used not only in a vacuum but in liquid environments [5] and has become an effective tool in electrochemistry. Rodahl *et al.* developed the QCM measurement system capable of simultaneously measuring the frequency, the absolute dissipation factor and the amplitude of oscillation of the quartz crystal [6]. This simultaneous measurement system provided information about the binding stiffness in adsorption studies. In the present study, the QCM measurement system has been developed using quartz crystals with a HAp coating. The adsorption of a bio-organic material onto HAp was investigated using the QCM system.

EXPERIMENTAL PROCEDURES

The AT-cut quartz crystal resonators with a fundamental resonance frequency of 6 MHz and equipped with a gold electrode from Hokuto Denko Corp. (Tokyo, Japan) were used as the sensors. The upper electrodes facing the liquid were coated with polycrystalline HAp thin films using the rf-sputtering method, followed by hydrothermal annealing at 433 K for 72 h [7]. The obtained layer was identified as a HAp single phase as shown in Fig.1. Scanning electron microphotographs of the HAp film surface are shown in Fig.2. The average thickness of the HAp films was approximately 2 μm. The HQ-200 liquid cell from Hokuto Denko Corp. was modified for the simultaneous

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measurements of the resonant frequency and dissipation factor. The liquid solution temperature stabilized at 37.0±0.1°C using water bath was injected into the cell at a constant flow rate by a tube pump.

The mass shift Δm on the HAp-coated electrode induces a linear decrease in the resonant frequency f when the mass is significantly small compared with the mass of the quartz crystal sensor. In this case, the Sauerbrey relation provides [8]:

$$\Delta m = -nC\Delta f \quad (1)$$

where Δf, C n are the resonant frequency shift, the mass sensitivity constant of the sensor, and the overtone number, respectively. Therefore, the small mass shift is easily detectable by precise measurement of the resonant frequency. In the present study, the QCM sensor was periodically disconnected from the driver of the signal generator.

The dissipation factor D was calculated from the measured decay curves of free oscillation after forced oscillation by the generator [6]. The decay curve is indicated as:

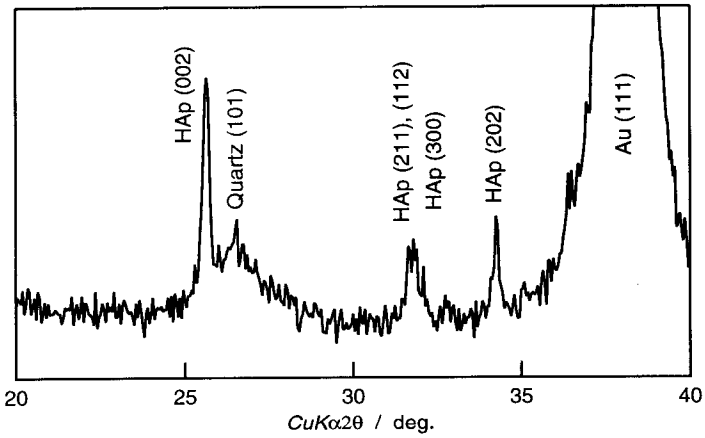

Fig.1 XRD pattern of HAp thin films coated on AT-cut quartz single crystal with gold electrode by rf-sputtering, followed by hydrothermal annealing at 433 K for 72 h.

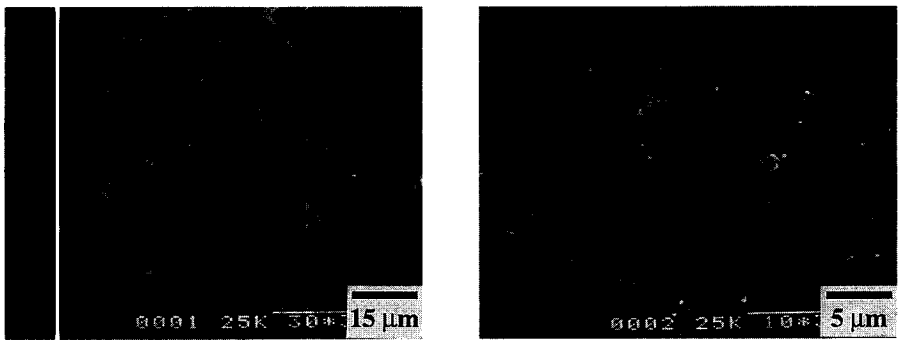

Fig.2 Scanning electron micrographs of HAp thin film surfaces coated on AT-cut quartz single crystal with gold electrode by rf-sputtering, followed by hydrothermal annealing at 433 K for 72 h.

$$y(t) = y(0)\exp(-t/\tau)\cos(\omega t + \varphi) \quad (2)$$

The t, $y(t)$, τ, ω and φ values are the time, the amplitude at t, the decay time, the angular frequency and the initial phase, respectively. The dissipation factor D is related to the decay time τ by:

$$D = 2/(\tau\omega) = 1/(\pi\tau f) \quad (3)$$

RESULTS AND DISCUSSION

The HAp composite QCM sensor placed in the liquid cell was washed with a continuous injection of extra pure water until the QCM system was stable, and then 0.2M *tris* HCl buffer (pH =9.0) was injected into the cell. The resonant frequency of the QCM sensor immersed in the buffer decreased compared with extra pure water as shown in Table 1. After the buffer immersion, the sensor was rinsed with 1000 ml of extra pure water for 1h. The resonant frequency did not revert but remained 50 Hz lower than that prior to the immersion. The irreversible shift in frequency suggested that the *tris* HCl buffer was strongly adsorbed onto the HAp surface. All of the D-shifts were positive and were not neglected. The positive shift showed that the energy loss of the QCM increased and that the viscosity and elasticity of the substance on the sensor surface changed. However, the reason for the change is not fully understood.

Table 1. Resonant Frequencies and Dissipation Factors of QCM Sensor Immersed in Extra Pure Water and 0.2M *tris* HCl Buffer as well as Extra Pure Water Rinsing for 1h after Buffer Immersion.

	f / Hz	$\Delta f = f - f_{aq}$ / Hz	D / 10^{-3}	$\Delta D = D - D_{aq}$ / 10^{-3}
Extra Pure Water	f_{aq} = 5962399	0	D_{aq} = 0.740	0
0.2M *tris* HCl Buffer (pH = 9.0)	f_b = 5962339	Δf_b = -60	D_b = 0.816	ΔD_b = 0.076
Rinsing with Extra Pure Water for 1 h after tris HCl Buffer Immersion	f_{aq}' = 5962349	$\Delta f_{aq}'$ = -50	D_{aq}' = 3.63	$\Delta D_{aq}'$ = 2.89

The decay curves for the HAp composite QCM sensor in 0.2M *tris* HCl buffer (pH = 9.0) and bovine serum albumin (BSA) solution are shown in Fig. 3. The BSA solution was 0.2M *tris* HCl buffer (pH = 9.0) with 1.0 μM of BSA (m_w = 66 kD, isoelectric point = 4.7- 4.9) provided by Calzyme Lab., Inc. (CA, USA). The temperature of the liquid solution was stable at 37.0±0.1°C. The BSA solution was injected at the flow rate of approximately 0.1 ml/s after the resonant frequency and dissipation factor were saturated and reached f_b = 5962444 Hz and D_b = 8.09×10^{-4} , respectively, due to the adsorption of the *tris* HCl buffer. The resonant frequency shifted negatively and became saturated at $\Delta f_{BSA} = f_{BSA} - f_b$ = -5 Hz less than 1 s after the start of the BSA solution injection. The dissipation factor linearly decreased and reached a saturation value of D_{BSA} = 6.50×10^{-4} at 30 min after the start of the BSA injection. Krim *et al.* reported that in a vacuum the slide between the QCM sensor surface and a molecular thin film adsorbed on the sensor surface increased the dissipation value [9]. Höök *et al.* showed using a QCM that in a liquid phase the dissipation factors were increased in the case of adsorption between the methyl terminated thiol and several kinds of proteins [10]. In the present study, on the contrary, the decline in the dissipation factor suggested that the linking strength of HAp-BSA, BSA-BSA and/or BSA-liquid were slowly changed. The significant decrease in the dissipation factor was attributed to the unique adsorbability of HAp with regard to proteins.

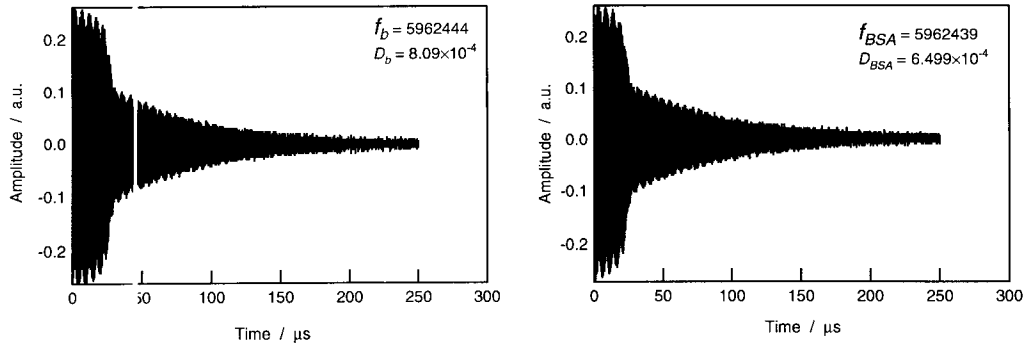

Fig.3 Decay curves for HAp composite QCM sensor immersed in 0.2M *tris* HCl buffer at *pH* = 9.0 (left) and bovine serum albumin solution diluted with 0.2M *tris* HCl buffer at *pH* = 9.0.

CONCLUSION

The HAP composite QCM sensor found that BSA was adsorbed onto the HAp surface and promptly reached saturation and showed that the dissipation factor decreased with a different time dependency from the mass of the adsorbates. It was implied that the HAp-QCM endowed with the unique effectivity for the detection of bio-organic materials.

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